Yagnik Pandya

Senior Data Scientist | AI & GenAI Engineer

Data Science & AI leader with 7+ years of hands-on experience delivering machine learning, NLP, computer vision, and generative AI solutions. Deep domain experience across logistics, supply chain, media, transportation, healthcare, and consulting. Proven track record architecting enterprise-scale RAG systems, LLM-powered applications, and cloud-native MLOps pipelines that drive measurable business outcomes. Skilled at leading cross-functional teams, mentoring data professionals, and translating complex business problems into scalable AI products and operational processes.

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CORE COMPETENCIES

Machine Learning & Deep Learning | Generative AI & LLMs | RAG | Prompt Engineering | Computer Vision | Natural Language Processing (NLP) | Time-Series Forecasting | MLOps & Cloud AI (AWS, GCP) | Data Engineering & Big Data | Stakeholder Management | Business Impact Analysis | Cross-Functional Collaboration | Visualization

WORK EXPERIENCE

Data Scientist
Deloitte Consulting India Pvt Ltd

02/2024 - Present *Pune, Maharashtra*

Tags: Prompt Engineering, Speech-to-Text, LLM, NLP, AI-Enhanced Validation, Google Blockly, Low-code development, Visual Programming, Stakeholder Management, Monaco editor, Google Gemini, OpenAI Embeddings, DataFlow, GCP, BigQuery, Firestore, QGIS, Time-Series Forecasting

- Architected and deployed a low-code Business Rules Editor (BRE) for VA DBQs, converting manual JSON rule creation into a visual workflow—reducing logic authoring errors by ~70% and validation downtime by **~90%.**
- Led Caltrans GenAI project delivery: built a RAG knowledge system using OpenAI embeddings + ChromaDB indexing 30K+ transportation documents, improving information retrieval accuracy and analyst productivity.
- Designed predictive models to estimate traffic buildup and incident likelihood by analyzing historical roadway data. Created **time-dependent features** including shifted observations, moving averages, and time-of-day signals, and assessed forecast reliability using **MAE** and **RMSE**.
- Built Army Recruit Assist GenAI application using AWS Bedrock, Lambda, Connect, and DynamoDB to assist call center representatives.
- Built **time-series** models to forecast **hourly inbound call** demand by capturing intraday seasonality and temporal trends from historical call data, enabling improved workforce planning and integration with live call summarization workflows.
- Implemented iterative MLOps feedback loops using Firestore to capture 1,400+ multi-turn interactions for LLM improvement and version control.

TECHNICAL SKILLS

Languages & Core
Python, SQL, Pyspark, FastAPI, GIT

Cloud & MLOps
Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, Apache Airflow, Docker, AWS Bedrock, AWS Connect, Lambda, DynamoDB, Kinesis, MLOps, ***Infrastructure-as-Code (IaC)****: Terraform or CloudFormation*

ML/DL Frameworks & Libraries
Scikit-learn, Keras, TensorFlow, Pandas, Numpy, Seaborn, Matplotlib, NLTK, Pytorch, OpenCV, Selenium, Pillow, beautifulsoup, SqlAlchemy, Paramiko, Transformers, Video analytics,

Big Data & Pipelines
Apache Beam/ DataFlow, Google BigQuery, Amazon Kinesis, Hudi, Kafka, EMR

Database/ Server
My SQL, MongoDB, Postgres

Vector Databases
ChromaDB, FAISS

Gen AI & LLMs
Gemini, Anthropic Claude, RAG, Prompt Engineering, GPT-4/3.5, Llama, Mistral, BERT, open-source models, LangChain, LlamaIndex, Hugging Face Transformers

Visualization & Other Tools
Power BI, Looker Studio, Tableau, Excel, PowerPoint

EDUCATION & CERTIFICATIONS

B.Tech in Computer Science (2016)
R. K University - Rajkot

AWS Certified Machine Learning – Specialty
(12/2024 - 12/2027)

AlmaBetter Data Science Bootcamp

Machine Learning by Andrew Ng

Statistics for Data Science (Udemy)

Deep Learning Specialization (Andrew Ng)

WORK EXPERIENCE

Data Scientist
News Corp India

05/2022 - 02/2024 *Bengaluru, Karnataka*

Tags: GCP, Bigquery, Recommendation system, Collaborative filtering, Matrix Factorization, Content-based filtering, Trend prediction, Topic modelling, NLP, BQML, Churn Forecasting, Content personalisation, Audience Behavior Analysis, Airflow,

- Developed collaborative filtering and content-based recommendation models using BQML, increasing reader engagement by 12% and click-through rate by 2.3%.
- Designed and deployed a RAG-powered editorial writing assistant leveraging article history to generate content suggestions and structured summaries.
- Lead The Times team to develop a **time series forecasting** model to identify the number of newspaper prints for each day reducing the loss of printing by **20%**.
- Produced dashboard ecosystems (Looker Studio) to monitor editorial performance, user consumption patterns, and ML model KPIs.

Data Science Engineer
BluePi Consulting Pvt. Ltd.

04/2021 - 05/2022 *Gurugram, Haryana*

Tags: CNN, Image detection, Image Classification, Open CV, Scrapping, Image enhancing, Image preprocessing, Resnet 50, Inception V3, MirNet, AWS SageMaker, Selenium, pixel normalization, Image augmentation, modeldb, pysftp, Kubernetes, EMR, airflow

- Developed axle-based vehicle classification model using MIRNet preprocessing achieving 85% accuracy; deployed using AWS SageMaker.
- Built cloud-native S3 Data Lake with Kafka GoldenGate streaming and incremental Hudi table pipelines for large-scale processing.
- Automated Selenium-based ingestion pipelines integrated with Apache Superset dashboards, eliminating repetitive reporting workload (~10+ hours/week).
- Operationalized ML workflows using Airflow on Kubernetes clusters for scalable and repeatable model deployment.

Operations Analyst
Samsara Shipping

09/2018 - 06/2020 *Pipavav Port, Gujarat*

Tags: Shipping, Logistics, Supply chain, Inventory Management, Time Series Analysis, Regression, Forecasting, Excel, Visualization, Time Series split, Grid Search, Performance Tuning

- Developed **vessel throughput and capacity forecasting models** using XGBoost (R^2 = 0.82), enabling **resource planning, berth allocation, and cost-efficient operations**.
- Performed feature engineering (lag variables, EWMA smoothing, differencing) and applied temporal cross-validation to enhance predictive robustness.
- Built **operations and supply chain KPI dashboards** tracking **turnaround time, asset utilization, throughput variance, and service levels** to support leadership decision-making.
- Collaborated with operations and vendor teams to analyze **cost drivers, demand variability, and supply risks** , supporting **data-driven sourcing and operational optimization decisions** .

RECOGNITION AT WORK

Outstanding Award (02/2025)

Spot Award (06/2024)

Spot Award (11/2024)

Best Team Award (06/2023)

ADDITIONAL PROJECTS

Sales Prediction of a Major Store Chain
AlmaBetter Verified Project

- Worked to Forecast Sales for a major drug store chain (Rossmann Store), and built a Machine Learning solution with Meta Regressor in the stacked ensembles model as XG Boost on base models such as Random Forest, Light GBM, and Decision Trees giving the lowest **RMSPE of 9.2%** as the final solution.
- **Engineered features include - days since the last promotion, days to next holiday, lag of sales, lag of customers** among others, and applied isolation forest to treat outliers and prepare data for further modeling by scaling it.
- Utilized analytical and visual expertise to provide insights and proposals to support sales improvement include Time Series Analysis .
- U s e d unsupervised learning using **PCA** for dimensionality reduction of highly correalted features and used **Bayesian Optimization, Grid Search to tune the hyperparameters.**

Stock Price Prediction
Personal Project

- Developed a robust stock price prediction model using historical data from Yahoo Finance.
- Integrated a Postgres database to efficiently manage and organize extensive stock records and manage ETL processes
- Engineered insightful features such as **Simple moving average (SMA), Exponential moving average (EMA)** for different periods of time, technical indicators, and volatility measures from raw data which includes open price, high price, low price, close price, and volume.
- Employed a suite of models including Linear Regression, Random Forest, and XGBoost, with rigorous evaluation. **The XGBoost model** emerged as optimal with an impressive **R2 score of 0.85**, demonstrating robust predictive capability.

Bike Demand Prediction
Analytics Vidhya Hackathon

- Forecasting hourly bike demand using historical usage data and temporal patterns, and built a **machine learning–based time-series solution** with **LightGBM as the best-performing model** .
- Engineered time-dependent features including **hour-of-day, day-of-week, lagged demand values, rolling aggregates, and weather-related indicators** , and applied appropriate data preprocessing to improve model stability and generalization.
- Evaluated model performance using standard forecasting metrics and iteratively tuned hyperparameters to optimize accuracy. The solution secured a **Top-200 rank in the Analytics Vidhya hackathon** , validating model effectiveness against a competitive benchmark.

CONTRIBUTIONS

Mentored junior data scientists and analysts in ML, problem framing, and deployment practices.

Conducted mock interviews and resume guidance sessions for early-career data professionals

Active contributor on Medium, Analytics Vidhya & StackOverflow (sharing tutorials and DS case studies).

Built Kaggle competition notebooks and RAG mini-projects for experimentation and continuous learning.

INTERESTS

Music | Cricket | Chess